

QUESTION OF THE WEEK 7
MATHEMATICS
GRADE XI

1. Find from definition the derivatives of the following functions:

- a) $\sin(3 - x)$
- b) $\sqrt{\tan x}$
- c) $\sqrt{\cot x}$
- d) $\sec(ax + b)$

2. Find $\frac{dy}{dx}$ when:

- a) $(x + y)^5 = x^3 y^2$
- b) $\sqrt{x} + \sqrt{y} = \sqrt{xy}$
- c) $x^2 + y^2 = (x + y)^2$
- d) $\sin(x + y) + \sin(x - y) = 1$
- e) $\cos(x + y) = \tan xy$

3. Find $\frac{dy}{dx}$ when:

- a) $y = ax^2 + bt + c$ and $t = \sqrt{x^2 + 1}$
- b) $y = 3\theta^2 + 4\theta + 5$ and $\theta = 3x^2 + 4x + 5$
- c) $y = at^3$ and $x = bt^2$
- d) $y = \frac{a}{t}$ and $x = \frac{b}{t^3}$

4.

- a) Define complex number. How does it differ from real number?
- b) Define absolute value of complex number with an example. Also, write its geometrical meaning.
- c) Define conjugate of complex number with an example. Also, give its geometrical interpretation.
- d) Write the properties of absolute value of complex number.
- e) Write the properties of conjugate of complex number.

5. Find the square roots of:

- a) $3 + 4i$
- b) $\frac{7+12i}{3-4i}$
- c) $\frac{1+i}{1-i}$